

UNIT I HARMONIC OSCILLATIONS

Introduction to harmonic oscillators

To and fro motion of a particle about the equilibrium position is known as oscillations.

Any motion which repeats itself after equal intervals of time is called a periodic motion or harmonic motion.

Ex: Spin of earth, the motion of a satellite around a planet, moon around the earth etc.

There are two types of oscillators

- i) Mechanical oscillators
- ii) Electrical oscillators

The oscillations which are due to change in mechanical quantities such as displacement, velocity, acceleration are known as mechanical oscillations. These oscillations belong to mechanical oscillator.

Ex: Spin of earth, moon around the earth etc..

The oscillations which are due to electrical and magnetic field vectors in visible light, radio waves etc are known as electrical oscillations. These oscillations belong to electrical oscillators.

Ex: A tuned circuit and closed metal cavity oscillate electro magnetically.

Simple harmonic motion

The oscillatory motion in which the force is directly proportional to the displacement is called simple harmonic motion (S.H.M). The particle or body executing S.H.M is called a simple oscillator. Since force is proportional to the displacement, the acceleration is not constant but varies with time.

Simple harmonic oscillator:

When a particle or body moves such that its acceleration is always directed towards a fixed point and varies directly as its distance from that point, the particle or body is said to execute S.H.M. The particle or body executing simple harmonic motion is called a simple oscillator.

Equation of motion of a simple oscillator:

Equation of motion of a simple oscillator

Consider a particle p of mass 'm' executing S.H.M about equilibrium position 'o' along X-axis. Let x be the displacement of p from o at any instant, then the restoring force F acting upon p is given by

$$F \propto -x$$

$$F = -kx \text{ ----- (1)}$$

Where 'k' is proportionality factor which represents the force per unit displacement, -ve sign indicates force F is opposite to the displacement.

According to Newton second law of motion

$$F = \text{mass} \times \text{acceleration}$$

$$F = m \frac{d^2x}{dt^2} \text{ ----- (2)}$$

From equation (1) and (2)

$$m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \omega^2x = 0$$

This is known as the differential equation of simple harmonic oscillation.

Solution using complex exponential method

The simple harmonic oscillator is

$$\frac{d^2x}{dt^2} + \omega^2x = 0 \text{ ----- (1)}$$

$$\text{Where } \omega^2 = \frac{k}{m}$$

Let us assume a trial solution of the form

$$x = Ce^{\alpha t} \text{ ----- (2)}$$

Where 'C' and 'α' are arbitrary constants.

By differentiating we get

$$\frac{dx}{dt} = C\alpha e^{\alpha t}$$

$$\frac{d^2x}{dt^2} = C\alpha^2 e^{\alpha t} \text{ ----- (3)}$$

Sub (2),(3) in (1), we get

$$C\alpha^2 e^{\alpha t} + \omega^2 C e^{\alpha t} = 0$$

$$C e^{\alpha t} (\alpha^2 + \omega^2) = 0$$

$$(\alpha^2 + \omega^2) = 0$$

$$\alpha^2 = -\omega^2$$

$$\alpha = \pm i\omega$$

From eq(2)

$$x = C_1 e^{+i\omega t} + C_2 e^{-i\omega t}$$

Where C_1 and C_2 are arbitrary constants

$$x = C_1 [\cos \omega t + i \sin \omega t] + C_2 [\cos \omega t - i \sin \omega t]$$

$$x = (C_1 + C_2) \cos \omega t + i(C_1 - C_2) \sin \omega t$$

Let us put $C_1 + C_2 = a \sin \phi$ and $C_1 - C_2 = a \cos \phi$

Where a and ϕ are constants

$$x = a \sin \phi \cos \omega t + i a \cos \phi \sin \omega t$$

$$x = a \sin(\omega t + \phi) \text{ ----- (4)}$$

Equation (4) is the solution of simple harmonic motion.

Damped harmonic oscillator:

Definition: Damped harmonic motion is defined as decay or decrease of amplitude of motion with respect to time in presence of air or other medium. The oscillator which performs such type of motion is known as damped harmonic oscillator.

Ex: Motion of tuning fork, Motion of simple pendulum. etc.

Equation for damped harmonic oscillator:

The damped system is subjected to:

i) A **restoring force** which is proportional to displacement but oppositely directed. This is written as $-kx$ where k is force constant.

ii) **Frictional force:** Frictional force is proportional to velocity but oppositely directed, given by

$$f_2 \propto -\frac{dx}{dt}$$

$$f_2 = -r \left(\frac{dx}{dt} \right) \text{ ----- (2)} \quad \text{Where 'r' is frictional force per unit velocity.}$$

Since, $\text{force} = \text{mass} \times \text{acceleration} = m \frac{d^2x}{dt^2}$

Therefore, the equation of the motion of particle is given by

$$m \frac{d^2x}{dt^2} = -kx - r \left(\frac{dx}{dt} \right) \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{r}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

Or

$$\frac{d^2x}{dt^2} + 2b \frac{dx}{dt} + w^2 x = 0 \quad \dots \dots \dots (1)$$

Where

$$\frac{r}{m} = 2b \text{ and } \frac{k}{m} = w^2$$

Here b is damping constant, and this is a differential equation of damped harmonic motion.

Solution of the equation:

Equation (1) is a differential equation of second degree. Let its solution be $x = Ae^{\alpha t}$... (2) where A and α arbitrary constants.

Differentiating equation 2 with respect to t, we get

$$\frac{dx}{dt} = A\alpha e^{\alpha t} \quad \text{and} \quad \frac{d^2x}{dt^2} = A\alpha^2 e^{\alpha t}$$

Substituting these values in eq.(1), we have

$$A\alpha^2 e^{\alpha t} + 2b A\alpha e^{\alpha t} + w^2 A e^{\alpha t} = 0$$

$$A e^{\alpha t} (\alpha^2 + 2b\alpha + w^2) = 0$$

As

$$A e^{\alpha t} \neq 0 \quad \therefore (\alpha^2 + 2b\alpha + w^2) = 0$$

This gives

$$\alpha = -b \pm \sqrt{b^2 - w^2}$$

The general solution of eq. (1) is given by

$$x = A_1 e^{[-b + \sqrt{b^2 - w^2}]t} + A_2 e^{[-b - \sqrt{b^2 - w^2}]t} \dots \dots (3)$$

Where A_1 and A_2 are arbitrary constants.

Different damping conditions:**Case – I, Over damped motion:**

When $b^2 > w^2$. In this case $\sqrt{b^2 - w^2}$ is real & less than b. Now the powers in eq. (3) are both negative. Thus this displacement x consists of two terms, both dying off exponentially to zero without performing any oscillations as shown in figure.

The rate of decrease of displacement is governed by the term $[-b + \sqrt{b^2 - w^2}]t$ as the other term reduced to zero.

In this case, the body once displaced returns to its equilibrium position quite slowly without performing any oscillation. This type of motion is called as **over damped** or **dead beat**.

Ex: Motion of simple pendulum moving in thick oil
a dead beat moving coil galvanometer.

Case II. Critical damping:

When $b^2 = w^2$, if we put $b^2 = w^2$ in equation (3), this solution does not satisfy the differential equation (1).

Let us consider $\sqrt{b^2 - w^2}$ is not zero, but this is equal to a very small quantity 'h'.
i.e. $\sqrt{b^2 - w^2} = h \rightarrow 0$

Now equation (3) reduces to

$$x = A_1 e^{(-b+h)t} + A_2 e^{-(b-h)t}$$

$$= e^{-bt} [A_1 e^{ht} + A_2 e^{-ht}]$$

$$= e^{-bt} [A_1 (1 + ht + \dots) + A_2 (1 - ht + \dots)]$$

$$= e^{-bt} [(A_1 + A_2) + ht(A_1 - A_2) + \dots \dots]$$

$$x = e^{-bt}(p + qt) \dots \dots (4)$$

Where $p = (A_1 + A_2)$ and $q = (A_1 - A_2)$

Eq.(4) represents a possible form of solution .

In this equation as 't' increases, e^{-bt} decreases and $[p + q t]$ increases and the damping is said to be critical.

Thus in the case of critical damping, the motion just becomes a periodic or non oscillatory.

Ex: This type of motion is exhibited by many pointer instruments such as Ammeter, **voltmeter**, etc, in which the pointer moves to the correct position and comes to rest much rapidly without any oscillations.

Case III. Under –damped motion:

When $b^2 < w^2$. In this case $\sqrt{b^2 - w^2}$ is imaginary. Let us write $\sqrt{b^2 - w^2} = i\sqrt{w^2 - b^2} = i\beta$ where $\beta = \sqrt{(w^2 - b^2)}$ and $i = \sqrt{-1}$
now eq.(3) becomes $x = A_1 e^{(-b+i\beta)t} + A_2 e^{(-b-i\beta)t}$

$$\begin{aligned} &= e^{-bt} (A_1 e^{i\beta t} + A_2 e^{-i\beta t}) \\ &= e^{-bt} ((A_1 \cos \beta t + i \sin \beta t) + A_2 (\cos \beta t - i \sin \beta t)) \\ &= e^{-bt} ((A_1 + A_2) \cos \beta t + i(A_1 - A_2) \sin \beta t) \\ &= e^{-bt} (a \sin \phi \cos \beta t + a \cos \phi \sin \beta t) \end{aligned}$$

Where $a \sin \phi = (A_1 + A_2)$ and $a \cos \phi = i(A_1 - A_2)$

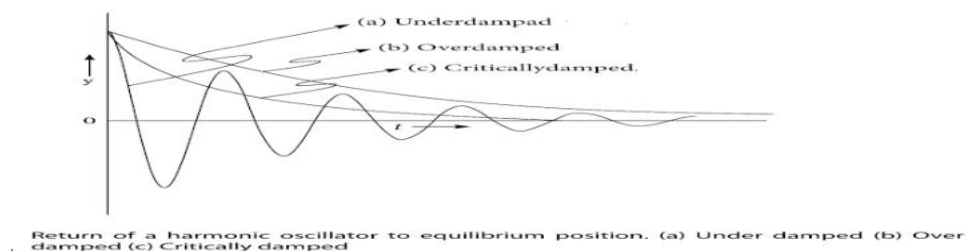
$$\begin{aligned} &= e^{-bt} a \sin(\beta t + \phi) \\ x &= a e^{-bt} \sin(\sqrt{(w^2 - b^2)}t + \phi) \end{aligned}$$

The above expression is the equation of motion of damped harmonic oscillator.

The amplitude of the motion continuously decreasing owing to the factor e^{-bt} which is called damping factor.

The factor $\sin(\sqrt{(w^2 - b^2)}t + \phi)$ varies between +1 to -1, therefore the amplitude also varies between $+a e^{-bt}$ and $-a e^{-bt}$. It is called under damped motion, b is called damping coefficient.

Ex: Motion of a pendulum in air, the electric oscillations of LCR circuit.



Energy decay in damped harmonic oscillator (Energy dissipation):

Whenever the system is set into oscillations, its motion is opposed by frictional (damping) forces due to air resistance. The work done against these forces is dissipated out in the form of heat. So the mechanical energy of the system continuously decreases with time and amplitude of oscillation gradually decays to zero.

The displacement of a damped harmonic oscillator at any time is given by

$$x = ae^{-bt} \sin \left(\sqrt{(w^2 - b^2)} t + \phi \right) \\ = ae^{-bt} \sin(w't + \phi) \dots \dots \dots (1)$$

$$\text{where } w' = \sqrt{w^2 - b^2}$$

$$\text{Then velocity, } \left(\frac{dx}{dt} \right) = -abe^{-bt} \sin(w't + \phi) + aw'e^{-bt} \cos(w't + \phi) \dots \dots \dots (2)$$

If damping is very small i.e., $w \gg b$. Hence the term $-abe^{-bt} \sin(w't + \phi)$ can be neglected in comparison with the term $aw'e^{-bt} \cos(w't + \phi)$.

$$\text{Then } \frac{dx}{dt} = aw'e^{-bt} \cos(w't + \phi) \dots \dots \dots (3)$$

The mechanical energy E of oscillator is given by

$$E = \text{Kinetic energy} + \text{Potential energy} \\ = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 + \frac{1}{2} \mu x^2 \dots \dots \dots (4)$$

Substituting the value of $\frac{dx}{dt}$ and x from equations (3) and (1) in eq.n (4),

$$E = \frac{1}{2} ma^2 w'^2 e^{-2bt} \cos^2(w't + \phi) + \frac{1}{2} \mu a^2 w'^2 e^{-2bt} \sin^2(w't + \phi) \\ = \frac{1}{2} ma^2 \frac{\mu}{m} e^{-2bt} \cos^2(w't + \phi) + \frac{1}{2} \mu a^2 e^{-2bt} \sin^2(w't + \phi) \\ \because w' = \sqrt{w^2 - b^2} \approx w = \sqrt{\left(\frac{\mu}{m} \right)} \\ = \frac{1}{2} a^2 \mu e^{-2bt} (\cos^2(w't + \phi) + \sin^2(w't + \phi)) \\ = \frac{1}{2} a^2 \mu e^{-2bt} \\ = \frac{1}{2} a^2 m w^2 e^{-2bt} \\ E = \frac{1}{2} a^2 \mu e^{-2bt}$$

This shows that the energy of oscillator decreases with time.

Quality factor(qualitative):

The quality factor is defined as 2π times the ratio of the energy stored in the system to the energy lost per period.

$$Q = 2\pi \frac{\text{energy stored in system}}{\text{energy lost per period}} = 2\pi \frac{E}{PT}$$

Where P is power dissipated and T is periodic time. We know that $P = \frac{E}{\tau}$, where τ is relaxation time. So

$$Q = 2\pi \frac{E}{\frac{E}{\tau} T} = \frac{2\pi\tau}{T} = w\tau$$

Where $w = \frac{2\pi}{T}$ = angular frequency.

Forced damped harmonic vibrations

Forced vibrations can be defined as the vibrations in which the body vibrates with a frequency other than its natural frequency under the action of an external periodic force. Ex: consider the vibrations of bridge under the influence of marching solders, vibrations of a tuning fork when exposed to the periodic force of sound waves.

Equation of motion of forced harmonic oscillator

The forces acted upon the particle are:

- i) A restoring force

Restoring force is proportional to the displacement but oppositely directed, given by

$$f_1 \propto -x$$

$$f_1 = -kx \text{ ----- (1)}$$

Where 'k' is known as the force constant.

- ii) Frictional force

Frictional force is proportional to velocity but oppositely directed, given by

$$f_2 \propto -\frac{dx}{dt}$$

$$f_2 = -r \left(\frac{dx}{dt} \right) \text{ ----- (2)}$$

Where 'r' is frictional force per unit velocity.

- iii) The external periodic force

It is represented by $f_3 = F \sin pt$ ----- (3)

Where F is the maximum value of this force

So the total force acting on the particle is given by

$$\text{Total force} = f_1 + f_2 + f_3$$

From (1),(2) and (3)

$$\text{Force}(F) = -kx + -r \left(\frac{dx}{dt} \right) + F \sin pt$$

According to Newton second law of motion

F = mass × acceleration

$$F = m \frac{d^2x}{dt^2}$$

Hence

$$m \frac{d^2x}{dt^2} = -kx - r \left(\frac{dx}{dt} \right) + F \sin pt$$

$$m \frac{d^2x}{dt^2} + kx + r \left(\frac{dx}{dt} \right) = F \sin pt$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x + \frac{r}{m}\left(\frac{dx}{dt}\right) = \frac{F}{m}\sin pt$$

$$\frac{d^2x}{dt^2} + 2b\left(\frac{dx}{dt}\right) + \omega^2x = f\sin pt \text{ ----- (4)}$$

Where $\frac{r}{m} = 2b$ and $\frac{k}{m} = \omega^2$ $\frac{F}{m} = f$

Equation (4) is the differential equation of motion of the particle.

Solution of equation of forced oscillations

In this case, when the steady state is setup, the particle vibrates with the frequency of applied force and not with its own natural frequency.

The solution of differential equation (4) must be of the type

$$x = A \sin(pt - \theta) \text{ ----- (5)}$$

Where A is the steady amplitude of vibration and θ is the angle by which the displacement x lags behind the applied force $F \sin pt$. A and θ are arbitrary constants.

Differentiating equation (5) we get

$$\frac{dx}{dt} = Ap \cos(pt - \theta) \text{ ----- (6)}$$

$$\text{And } \frac{d^2x}{dt^2} = -Ap^2 \sin(pt - \theta) \text{ ----- (7)}$$

Sub (5),(6),(7) in (4), we get

$$-Ap^2 \sin(pt - \theta) + 2bAp \cos(pt - \theta) + \omega^2 A \sin(pt - \theta) = f \sin pt$$

$$A \sin(pt - \theta) (\omega^2 - p^2) + 2bAp \cos(pt - \theta) = f \sin(pt - \theta + \theta)$$

$$A \sin(pt - \theta) (\omega^2 - p^2) + 2bAp \cos(pt - \theta) = f \sin\{(pt - \theta) + \theta\}$$

$$A \sin(pt - \theta) (\omega^2 - p^2) + 2bAp \cos(pt - \theta) = f \sin(pt - \theta) \cos \theta + f \cos(pt - \theta) \sin \theta$$

Comparing the coefficients of $\sin(pt - \theta)$ and $\cos(pt - \theta)$ on b/s we get

$$A(\omega^2 - p^2) = f \cos \theta \text{ ----- (8)}$$

$$2bAp = f \sin \theta \text{ ----- (9)}$$

Square and add equations (8) and (9), we get

$$A^2(\omega^2 - p^2)^2 + 4b^2A^2p^2 = f^2$$

$$A = \frac{f}{\sqrt{(\omega^2 - p^2)^2 + 4b^2A^2p^2}} \text{ ----- (10)}$$

Where A is amplitude of forced vibrations.

Divide (9) by (8), we get

$$\frac{f \sin \theta}{f \cos \theta} = \frac{2bAp}{A(\omega^2 - p^2)}$$

$$\tan \theta = \frac{2bp}{(\omega^2 - p^2)}$$

$$\theta = \tan^{-1} \left(\frac{2bp}{(\omega^2 - p^2)} \right)$$

This is called phase of vibrations.

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